

WEEK 1: INTEGRATIVE PHYSIOLOGY

CELLULAR SIGNALING AND SIGNAL TRANSDUCTION

CHEMICAL MESSENGERS

- Bind to protein receptors on the surface of the cell, in the cytoplasm or the nucleus, which triggers sequences of intracellular changes that produce their physiologic effects
- Chemical substances which carry messages or signals from signaling cells to target cells
- E.g., amines, amino acids, steroids, polypeptides
- **Kinds:**
 - Endocrine messengers
 - "Classical" hormones
 - Hormone is synthesized *by endocrine glands* and **transported by blood** to the target organs/tissues
 - E.g., growth hormone and insulin
 - Paracrine messengers
 - **Diffuse through interstitial fluid** from the control cells to the target cells
 - E.g., prostaglandin and histamine
 - Type: Juxtacrine messengers (local hormones) - directly enter neighboring target cells through the gap junctions
 - Autocrine messengers
 - AKA intracellular chemical mediators
 - Bind to receptors on the source cell—the same cell it was secreted from
 - Basically, what a signaling cell releases will also be used by that same exact cell
 - E.g., leukotrienes
 - Neurocrine messengers
 - AKA neural messengers
 - Neurotransmitters
 - Endogenous signaling molecule
 - E.g., ACh and dopamine
 - Basically, what happens at synaptic junction
 - Neurohormones
 - Released by nerve cells directly into the blood and transported to distant target cells
 - E.g., oxytocin, antidiuretic hormone, hypothalamic releasing hormone

- Basically, what the axon terminal releases

RECEPTORS

- Protein molecules on the sensing cells which sense chemical changes to activate response
- Can be found in the cell membrane, cytoplasm, and nucleus

TYPES OF INTRACELLULAR COMMUNICATION

- Transfer of information from one cell to another

1.) Gap junctions

- AKA intercellular connection
- Allows intercellular signaling molecules to diffuse from the cytoplasm between two cells
- Passes freely without ECF (since synaptic junctions are present)
- Permits the transfer of ions, sugars, amino acids
- Permits rapid propagation of electrical activity, necessary for cardiac and smooth muscle

2.) Neural communication

- Neurotransmitters: released at the synapse
- Fast communication (can be done within milliseconds). Thus, can travel long distances
- Effect limited to very specific (discrete) areas (ACh binding to Na voltage-gated channel)

3.) Endocrine communication

- Hormones are released into bloodstream/lymph, affects distant cells
- Slower = requires time (from secs to hrs) for diffusion and blood to flow to the target cell
- Effect is prolonged
- Effect is either limited to discrete or generalized (diffused) areas

4.) Paracrine communication

- Release of substances into the ECF
- Affects nearby cells. Thus, short-distance travel
 - Signaling cell to target cell

5.) Autocrine communication

- Release of substances through ECF
- Limited to the cell that secretes the hormone or itself

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- Signaling cell takes what it releases

CELL SIGNALING AND SIGNAL TRANSDUCTION

- Signaling molecule (e.g., hormone or neurotransmitter) binds to a receptor, that is either in the membrane, cytosol, or nucleus, to activate intracellular signaling protein which regulate the activity of one or more target proteins for cellular function

- Signaling molecule:

- Regulate cell growth, division, and differentiation
- Influence cellular metabolism
- Regulate activity of ion channels and transport protein intracellularly
- Control cytoskeleton-associated events (e.g., cell shape, division, migration, and cell-to-cell and cell-to-matrix adhesion)

PROCESS OF INTRACELLULAR COMMUNICATION

1.) Recognition

- Transmitter-receptor binding

- Begins by binding signal molecule to receptor at cell

- Regulation of receptors:

- Not static; increases or decreases in response to stimuli
- Change in physiologic condition -> change in receptor properties
 - Up-regulation - increase in the number of receptors; chemical messenger is deficient
 - Down-regulation - decrease in number of receptors; chemical messenger is in excess
 - Receptor-mediated endocytosis - sometimes responsible for down-regulation; once the ligand is binded to the receptor, it is taken into the cell via endocytosis
 - Desensitization - decrease in responsiveness; receptor inactivation
 - EXCEPTION: Angiotensin II increases the number of receptors

2.) Transduction

- Second messenger

- E.g., cAMP, IP3

- Signaling proteins interact and regulate the activity of target proteins
- Modulate cellular function
- Target proteins where second messenger is produced include:
 - Ion channels and other transport proteins
 - Metabolic enzymes
 - Cytoskeleton proteins
 - Gene regulatory proteins
 - Cell cycle protein

3.) Response

- Regulate cell growth, division, and differentiation

- Influence cellular metabolism

- Regulate activity of ion channels and transport protein intracellularly

- Control cytoskeleton-associated events (e.g., cell shape, division, migration, and cell-to-cell and cell-to-matrix adhesion)

TYPES OF RECEPTORS

1.) Ion channel-linked receptors

- AKA ligand gated ion channels or ionotropic receptors
- Direct and rapid synaptic signaling between electrically excitable cells
- Major site where anesthetic agents and ethanol have their effects
- Basically, a neurotransmitter binds to ion gated channels for it to open. Ion influx cause AP propagation, which will depolarize the postsynaptic cell membrane
 - Acetylcholine gated Na channel
 - Increased membrane permeability to Na
 - ACh receptors are important in the transmission of electrical signals between excitable cells

2.) G-Protein-coupled receptors (GPCR)

- AKA metabotropic receptors

- Passes through membrane 7 times

- AKA 7 transmembrane receptor, heptahelical receptor, 7-helix receptor, serpentine receptor
- May be stimulated or inhibited via ligand-bound receptors
- Interacts with different kind of effector proteins/enzymes causing cascade of events

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- Basically, GPCR receptors have alpha (site of ADP), beta, and gamma subunits that are bound intracellularly. Once a ligand is bound to the GPCR extracellularly, GDP will undergo phosphorylation to be GTP. Once this happens, alpha subunit detach from beta and gamma subunits, then the alpha subunit will attach to a target protein, which will bring rise to a second messenger. The second messenger carries a message for cellular processes. Once it's all done, alpha subunit will detach from the target protein via GTP-activated target protein enzyme, which will also act as an adhesion for alpha, beta, and gamma subunit to be attached once again, as well as hydrolysis from GTP to GDP.

3.) Enzyme-linked/catalytic receptors

- AKA kinase-linked receptors
- Acts immediately as an enzyme/closely associated enzyme when activated
- Pass through membrane only once
- Most are protein kinases or associated with it
- Basically, enzyme-linked receptors are inactive catalytic domains, containing two pieces of receptor with a space in between. Once a ligand is present extracellularly, it will bind the two pieces together, phosphorylating ADP to ATP immediately

4.) Nuclear receptors

- Found inside the cell cytosol, which then enters the nucleus when bonded with hormones
 - Hormones are adrenal, gonadal, steroid, thyroid, retinoid, and vitamin D
 - Progesterone, estrogen, thyroid - bound to the DNA of the nucleus even in the absence of the hormone
- Basically, an activated hormone receptor (either cytosol receptor or nuclear receptor) binds to a promoter sequence of the DNA called the hormone response element. This activates or represses transcription of specific chains and formation of mRNA for protein synthesis

SIGNAL TRANSDUCTION

- Process occurring within the cell that convert one kind of stimulus into another type
 - First messenger to second messenger
- Cells ability to change behavior in response to a receptor-ligand interaction

- Kinds of messenger:

- First messenger
 - Extracellular ligands
 - When bound to a receptor, second messenger is expected to occur
- Second messenger
 - Intracellular mediators
 - Relays signals from one location to another
 - Amplifies response through changes within the cell or cascade of events

GPCR SIGNAL TRANSDUCTION TYPES

1.) Adenylyl cyclase/cAMP-signal pathway (enzyme)

SECOND MESSENGER: cAMP

- Adenylyl cyclase
 - Very common downstream effector of G-protein
 - Facilitates conversion of ATP to cAMP
 - When its protein kinase A is activated, it phosphorylates protein to either stimulate or inhibit enzymes
- cAMP regulated some ion gated channels
- Basically, once the alpha subunit attaches to adenylyl cyclase, it could either go in two ways:
 - Alpha subunit binds to alpha s class - activates adenylyl cyclase, increases cAMP, activate protein kinase A
 - Alpha subunit binds to alpha i class (GiPCP) - inhibition of adenylyl cyclase, decreases cAMP, decreases protein kinase A

2.) Phosphodiesterase pathway

SECOND MESSENGER: cGMP

- Ligand: light
 - When light hits the photoreceptor, the light-absorbing retinal portion of the rhodopsin is activated
 - When this activates, it stimulates transducin (G-protein) which then activates cGMP phosphodiesterase
- Phosphodiesterase
 - Degradation of cGMP to GMP
 - **Reduction of cGMP closes Na channels**, which causes hyperpolarization of the photoreceptors
- Basically, when ligand (light) hits the transducin (receptor), alpha subunit dissociates with beta and

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gamma subunit. Alpha subunit then attaches to a protein effector called phosphodiesterase to convert cGMP to GMP. On the other hand, as the cGMP is broken down, cGMP-dependent channels (Na channels) close as well

3.) Phospholipase pathway

SECOND MESSENGERS: diacylglycerol (DAG) or inositol triphosphate (IP3)

- Phospholipase

- Stimulates phospholipase C (PLC) and it could either make DAG or IP3
 - IP3: binds to IP3 receptors in smooth ER and mitochondria important for opening of Ca channels; important for muscle contraction
 - DAG: activates protein kinase C for cellular response

- Phospholipase A2 (PLA2) releases arachidonic acid

- If released from cells, necessary for inflammation
- If retained inside, forms second messenger intracellularly

4.) Arachidonic acid (ARA) pathway

3 PATHWAYS:

1.) Metabolism of ARA to prostaglandin, thromboxane, and prostacyclins via cyclooxygenase

- Prostaglandin
 - Platelet aggregation; constriction of airways; induce inflammation
- Thromboxane
 - Platelet aggregation
 - Constrict blood vessels
- Prostacyclins
 - Inhibit platelet aggregation

2.) Metabolism of ARA to leukotrienes via

5-lipoxygenase

- Leukotrienes
 - For allergic and inflammatory reactions that cause asthma, rheumatoid arthritis, and inflammatory bowel disease

3.) Metabolism of ARA of HETE and EET via epoxigenase (CYP450)

- HETE and EET
 - Increase release of Ca from ER
 - Stimulate cell proliferation

CLASSES OF ENZYME/CATALYTIC RECEPTOR-LINKED

1.) Atrial natriuretic peptide (ANP) receptor

- Dilates blood vessels and stimulate Na secretion in urine
- Basically, a ligand binds to an extracellular receptor. Alpha, beta, and gamma subunits dissociate, and GDP to GTP phosphorylation. Alpha subunit binds to target protein, which causes conformational change. After this, it activates guanylyl cyclase, turning GTP from alpha subunit to cGMP. cGMP will then activate cGMP dependent protein kinase which phosphorylates proteins

2.) Receptor guanylyl cyclase

- Nitric oxide, NO receptor
- Relaxes arteriole smooth muscles
 - Activates soluble RGC. Dissociation of alpha, beta, and gamma subunit, and GDP to GTP phosphorylation. Alpha subunit binds to target protein, which activates guanylyl cyclase, turning GTP from alpha subunit to cGMP. cGMP will then act on nitroglycerine to increase nitric oxide production to relax smooth muscles and is necessary for treating chest pains due to inadequate blood flow to the heart

3.) Receptor serine/threonine kinase

- It phosphorylate serine/threonine that are found in proteins
- Composed of two subunits
- Basically, for example, beta transforming growth factor (TGF-β) binds to type II subunit and induces phosphorylation of type I subunit. This activity activates downstream substrates and regulatory proteins, which causes transcription of target genes that function in differentiation chemotaxis, proliferation, and activation of many immune cells

4.) Receptor Tyrosine Kinase

- High affinity cell surface receptors
- Composed of 2-alpha and 2-beta subunits
- Examples: insulin, nerve growth factor, epidermal growth factor
 - Insulin
 - Basically, insulin in alpha subunits causes conformational change which

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facilitates interaction between the alpha and beta pairs. Binding of insulin to receptors causes autophosphorylation of beta subunits. The activated receptor will then phosphorylate cytoplasmic protein to initiate cellular effects to control blood glucose levels, synthesis of lipids, and regulation of enzymatic activity

- NGF
 - Growth, maintenance, proliferation, and survival of certain target neurons
 - Facilitate dimerization and activation of tyrosine kinase activity
- EGF
 - Inhibition of gastric acid secretion, fetal growth, and development and neuromodulation in the CNS

5.) Tyrosine kinase associated receptors

- Interact with cytosolic tyrosine kinases (JAK)
- Examples: interleukins-6, erythropoietin
- Basically, when ligand binds to a receptor, its subunit assemble into homodimers, and heterodimers. This assembly enhances binding of tyrosine kinase (JAK) which increases kinase activities and thereby phosphorylates JAK and its receptor

DIRECT ACTIVATION OF PROTEIN TRANSCRIPTION

Nuclear receptor-linked signal transduction pathway: regulation of gene

- Family of intracellular ligand-regulated transcription factors
 - Ligands should be lipid-soluble as they need to cross the plasma membrane
 - Interact with nuclear receptors inside the cell
 - Usually steroid hormones (estrogen) or other lipid-soluble signals (retinoic acid)
 - Steroid hormone receptors - bind retinoic acid, thyroid hormones, and vitamin D
 - When ligand binds into these receptors, it activates transcription factors that bind DNA and regulate the expression of gene

NUCLEAR RECEPTOR TYPES

1.) Type I - cytoplasmic receptors

- Bound to cytoplasm by chaperone proteins
- Basically, when hormones go to this, it results in a conformation change that causes chaperones to dissociate from the receptor thereby uncovering nuclear localization motif revealing hormone receptors from the nucleus. Glucocorticoids and mineralocorticoids bind to the hormone receptor to induce action. Additionally, glucocorticoids suppress transcription activators to regulate gene expression

2.) Type II - nuclear receptors

- Located in the nucleus, bound to DNA
- Have hormone receptor
- Estrogen and progesterone are the hormones present in the nucleus
- Thyroid hormone and retinoid acid are also present but is bound to DNA

REGULATION OF TRANSCRIPTION

- After hormone binds to the receptor in the cytoplasm, or hormone receptor in the nucleus, it goes inside the nucleus to bind to the hormone response element (promoter). The action could either activate or inhibit gene transcription (either form mRNA or not) for protein synthesis

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SYNAPTIC & JUNCTIONAL TRANSMISSION

FUNCTIONS OF SYNAPTIC ELEMENTS

- **Dendrites** - major sites that receive electrical signals from other neurons
 - **Dendritic spine** - dynamic structure, and is important for learning motivation, and long-term memory
 - Local protein synthesis can occur in dendrites
- **Synaptic cleft** - 20-40 nm wide gap between presynaptic and postsynaptic neurons
- **Postsynaptic density** - thickened region full of receptors, enzymes, and binding proteins that ensures effective signaling
- Synaptic vesicles:
 - Types:
 - **Small clear vesicles** - ACh, glycine, GABA, glutamate
 - **Small dense-core vesicles** - catecholamines (dopamine, norepinephrine, epinephrine)
 - **Large dense-core vesicles** - neuropeptides
 - Neurotransmitter release:
 - **Active zones** - specialized membrane areas where vesicles release neurotransmitters
 - **Voltage-gated Ca channels** - near active zones (Ca entry triggers exocytosis)
 - **Neurexin-neuroligin interactions** - adhesion molecules that align pre- and postsynaptic sites, ensuring synaptic specificity
 - Toxins & clinical relevance
 - Some bacterial toxins block neurotransmitter release by cutting SNARE proteins
 - **Botulinum toxin** - prevent ACh release at neuromuscular junction, causing paralysis
 - **Tetanus toxin** - blocks inhibitory transmitter release in CNS, causing spasms

ELECTRICAL EVENTS IN POSTSYNAPTIC NEURONS

EXCITATORY AND INHIBITORY POSTSYNAPTIC POTENTIALS

- When a sensory neuron (dorsal root afferent) is stimulated, there's a short delay (-0.5 ms) before the postsynaptic neuron responds
 - This delay is due to the time needed for neurotransmitter release and receptor activation
 - Because of this, pathways with many synapses conduct more slowly than those with few synapses
 - Used to determine if a reflex is monosynaptic or polysynaptic
- Excitatory postsynaptic potential (EPSP)
 - A single stimulus does not trigger a full AP
 - Causes small depolarization of the postsynaptic membrane
 - Begin -0.5 ms after input
 - Peak at 1-1.5 ms, then fade
 - **Caused by Na or Ca influx**
 - One EPSP alone is too weak, but multiple EPSPs can summate to trigger AP
- Inhibitory postsynaptic potential (IPSP)
 - Other inputs can cause hyperpolarization instead of depolarization
 - Also peak at 1-1.5 ms after stimulus
 - **Caused by -Cl influx efflux or K efflux**
 - Makes the membrane potential move farther from threshold - less excitable
 - More excitatory input is required to trigger firing
- Ion channel basis
 - EPSPs - opening of Na or Ca channels - inward positive current
 - IPSPs - opening of -Cl or K channels - hyperpolarization
 - IPSPs can disappear at the chloride equilibrium potential and even reverse direction depending on the membrane voltage

SLOW POSTSYNAPTIC TERMINALS

- Occurs: autonomic ganglia, cardiac muscle, smooth muscle, cortical neurons
- Latency: 100-500 ms (slower than fast PSPs)
- Last several seconds (instead of ms)
- Slow EPSPs
 - Less K efflux -> membrane stays more positive -> depolarization -> higher excitability

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- Slower IPSP

- More K efflux -> membrane becomes more negative -> hyperpolarization -> lower excitability

GENERATION OF AN ACTION POTENTIAL IN THE POSTSYNAPTIC NEURON

- Neurons constantly receives EPSPs and IPSPs signals

- Add together algebraically (depolarizations + hyperpolarizations)
- Soma acts as an integrator - decides whether the combined effect is enough for threshold

- Decisions happen at axon hillock and initial segment (unmyelinated part beyond soma) = lowest threshold region

- Trigger zone where action potential usually begins

- If it reaches threshold

- Action potential propagates down the axon toward terminals
- Can travel backward into the soma (retrograde firing) - resets and help prevent leftover EPSPs & IPSPs to interfere with new signals

TEMPORAL & SPATIAL SUMMATION POSTSYNAPTIC POTENTIALS

- Temporal summation

- Longer time, more opportunity for overlap
- If a second EPSP arrives before the first decays, they add up
 - Push neuron to threshold

- Spatial summation

- How far depolarizing current can spread along the neuron before it fades away
- Longer length, depolarization spreads farther with less decay
- If presynaptic neurons from different locations all arrive with enough strength, they summate

POSTSYNAPTIC INHIBITION

- Inhibitory transmitter, such as GABA or glycine, is released from a presynaptic nerve terminal onto the postsynaptic neuron to induce an IPSP

- E.g., Afferent fibers from the muscle spindles (stretch receptors) in skeletal muscle

- When the muscle is stretched, it fires and excites the motor neuron of the same muscle via EPSPs, causing contraction
- At the same time, it activates inhibitory interneuron to release GABA or glycine onto the motor neurons of the antagonistic muscle, producing IPSPs
- Result: antagonistic muscle is inhibited (relaxes), while agonist contracts

PRESYNAPTIC INHIBITION & FACILITATION

- Happens at axoaxonal synapses

- Increased -Cl influx - smaller action potential at the terminal
- K channels open - K efflux, shorter depolarization
- Both effects - less Ca entry, less neurotransmitter released

- Key neurotransmitter: GABA

- Gaba a - Increase -Cl influx (fast inhibition)
- Gaba b - Increase K efflux (slower inhibition)

- Prolonged action potential in the terminal causes the Ca to stay open longer. Thus, more neurotransmitters are released

NEUROMUSCULAR TRANSMISSION

NEUROMUSCULAR JUNCTION

- Presynaptic terminal

- As the axon approaches the muscle, it loses its myelin sheath and branches into terminal boutons (synaptic terminals)
- The terminals contain vesicles filled with ACh
- Mitochondria is also present for energy

- Synaptic cleft

- Tiny space between the presynaptic and muscle membrane
- Contains acetylcholinesterase (AChE)

- Postsynaptic terminal

- Special region of the muscle fiber membrane (sarcolemma) directly beneath the terminal buttons
- Has junctional folds - deep invaginations that increase surface area
 - rich in Nicotinic ACh receptors (nAChRs), which responds to ACh

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SEQUENCE OF EVENTS DURING TRANSMISSION

Action potential travels down the motor neuron to the axon terminal. Causes depolarization, which opens voltage-gated Ca channels, and Ca ions enter synaptic bouton. Diffusion of Ca triggers vesicle fusion with the presynaptic membrane to release ACh into the synaptic cleft. ACh diffuses across and binds to nicotinic cholinergic receptors on the motor endplate, which are ligand-gated Na/K ion channels. Once ACh binds, Na influx and K efflux happens. Domination of Na influx causes local depolarization (endplate potential). The EPP spreads to the nearby muscle membrane (sarcolemma). If the depolarization reaches threshold, voltage-gated Na channels open, causing self-propagation of AP across muscle fiber. The AP dives into T-tubules to trigger SR to release Ca. Ca binds to troponin, which moves tropomyosin out. Actin and myosin can now slide. Thus, muscle contracts. After contraction, the system must reset to prevent continuous firing. AChE enzyme in the synaptic cleft breaks down ACh into acetate + choline. Choline is recycled back into the presynaptic terminal to make ACh. No ACh means no more Na influx. Thus, muscle relaxes

QUANTAL RELEASE OF TRANSMITTER

- Quanta of ACh
 - Isn't release in a continuous flow
 - Quanta - packets filled with ACh, each corresponding to the contents of one synaptic vesicle
 - 1 vesicle = ~10,000 molecules of ACh
- Miniature endplate potentials (MEPPs)
 - Even at rest, the terminal randomly releases single vesicles of ACh about 0.5 mV
 - Do not trigger contraction, but they're evidence of vesicle-based (quantal) release
 - More Ca = more vesicle release
 - More Mg = less vesicle release
- Large endplate potential (EPP)
 - When an action potential reaches terminal, hundreds (minimum of 60) to thousands vesicles release ACh simultaneously
 - Terminated by acetylcholinesterase
 - Strong enough to reach threshold
- Diseases:
 - Myasthenia gravis

- Antibodies attack nicotinic ACh receptors at the muscle endplate
- Lambert-Eaton Myasthenic Syndrome (LEMS)
 - Antibodies attack presynaptic Ca channels

NERVE ENDINGS IN SMOOTH AND CARDIAC MUSCLES

- Postganglionic autonomic nerve fibers
 - Nerve fibers from the ANS that branch extensively and come into close contact with smooth muscle cells in various organs
 - Do not form specialized endplates
 - Contains clear vesicles with ACh or dense-core vesicles with norepinephrine
 - Varicosities - studded with bead-like enlargements, each containing synaptic vesicles
 - Noradrenergic - have 20,000
 - Synapse en passant
 - Axon forms multiple contacts as it passes along the surface of the cells
- Neuroeffector junctions in the heart
 - Cholinergic and noradrenergic fibers terminate on the sinoatrial node, atrioventricular node, and bundle of His
 - Nora also innervate ventricular muscle

JUNCTIONAL POTENTIALS

- Excitatory junction potentials (EJPs)
 - Caused by stimulation of noradrenergic nerves in the smooth muscles produces discrete partial depolarization
 - EJPs can summate, increasing depolarizing effect
 - Can also be seen in tissues excited by cholinergic discharges
- Inhibitory junction potentials (IJPs)
 - In tissues where noradrenergic activity is inhibitory, nerve stimulation produces hyperpolarizing potentials
- EJPs and IJPs spread through smooth muscle fibers in an electronic fashion

AXONAL INJURY & DENERVATION SUPERSENSITIVITY

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- Orthograde (Wallerian) degeneration
 - Occurs distal to the point of injury, interrupting neural transmission
 - Axonal membrane breaks down, and the myelin sheath degenerates
- Retrograde degeneration
 - Proximal axon may also degenerate and die
 - Cell body swells, nucleus is displaced to the side (eccentric nucleus), and fragmentation of RER
- Regeneration and sprouting
 - Regenerative sprouting - injured nerves attempt regrowth
 - Sometimes axons successfully reconnect with their original targets, especially at the neuromuscular junction
 - Regeneration is usually limited because axons often get entangled in scar tissue at the injury site
 - Neutrophins - can improve regeneration
- Denervation hypersensitivity (supersensitivity)
 - When a motor nerve from skeletal muscle is cut and degenerates, the muscle becomes extremely sensitive to ACh
 - Also occurs in autonomic junctions
 - Skeletal muscle - gradually atrophies when denervated
 - Smooth muscle - does not atrophy, but becomes hyperresponsive to its neurotransmitter
 - Happens because of upregulation of receptors due to deficiency of neurotransmitters
 - Lack of reuptake of secreted neurotransmitters, prolonging their effect

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NEUROTRANSMITTERS AND NEUROMODULATORS

INTRODUCTION

- Major neurotransmitters discussed:
 - Excitatory amino acids
 - Inhibitory amino acids
 - Acetylcholine
 - Monoamines
 - Neuropeptides
- Common steps in neurotransmission:
 - Uptake of precursor into nerve terminal
 - Biosynthesis of neurotransmitter
 - Storage in synaptic vesicles
 - Release into synaptic cleft upon depolarization of presynaptic terminal
 - Binding to postsynaptic receptors
 - Termination of action via:
 - Diffusion away from synapse, reuptake into presynaptic terminal, enzymatic degradation

CHEMISTRY OF TRANSMITTERS

- **Neurotransmitters & enzymes** are concentrated in nerve endings (axon terminal), allowing efficient release and regulations

- Three main classes of transmitters:

- **Small-molecule transmitters**
 - *Amino acid*: glutamate (eventually turns to GABA), γ -aminobutyric acid (GABA), glycine
 - *ACh* (Acetyl + Choline)
 - *Monoamines*: dopamine, norepinephrine, epinephrine, serotonin
 - NE - found at locus coeruleus
 - Serotonin - found at raphe nuclei
 - Dopamine - found at substantia nigra
 - ACh - found at septal nuclei and nucleus basalis
- **Large-molecule transmitters (neuropeptides)**
 - Substance P, enkephalin, vasopressin
 - Frequently co-localized with small-molecule transmitters
- **Gas transmitters**
 - Nitric oxide and carbon monoxide

RECEPTORS

- Effect depends more on receptor type than on the mediator (neurotransmitter) itself
- Can be found in the cell membrane, cytoplasm, and nucleus
- **Receptors on both pre- and postsynaptic sites:**
 - **Autoreceptors** - localized on presynaptic endings; they inhibit further transmitter release
 - NE on α_2 -presynaptic receptors inhibits additional norepinephrine release
 - **Heteroreceptors** - ligand is different from transmitter of that neuron; neurotransmitter affects other neurotransmitter (not its own kind)
 - NE acts on a cholinergic nerve ending, inhibiting ACh release
- Some presynaptic receptors instead facilitate neurotransmitter release
- **Two major receptor families:**
 - Ionotropic (ligand-gated channels) - fast synaptic transmission (ms)
 - Neurotransmitter binds to ligand-gated channel, then channel opens for the ion to flow across membrane
 - Metabotropic (GPCRs) - activate 2nd messengers, modulate ion channels, slower but longer-lasting
 - Neurotransmitter binds to a receptor. the ADP attached intracellularly at the receptor will then undergo phosphorylation (ADP to ATP). This will cause the alpha subunits of G-protein to be separated from beta and gamma subunits. The alpha subunit of G-protein will then attach to the effector protein, releasing a second messenger. The second messengers will be the effector to cause actions throughout the membrane (e.g., opening of channels). The alpha subunit can detach from the effector protein if there is GTP-activated protein enzyme. This will help detach, as well as bind the alpha subunit together with beta and gamma subunit. Along with this, ATP will be hydrolyzed back to ADP.

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- Receptors are concentrated in clusters on the postsynaptic membrane, near nerve endings that release their neurotransmitters

- This localization is due to specific binding proteins that stabilize receptor placement

- With prolonged ligand exposure, receptor lose responsiveness

- Homologous desensitization - unresponsive to its own ligand, while sensitivity to other ligands remains
- Heterologous desensitization - unresponsive to multiple ligands, not just the one bound

REUPTAKE

- Rapid transport of neurotransmitters from synaptic cleft back into the cytoplasm of the nerve ending that released them

- Mediated by high-affinity, Na-dependent membrane transporters

- Example:

- Norepinephrine reuptake
 - After it is released into synaptic cleft, it is rapidly transported back into the sympathetic postganglionic nerve ending by the norepinephrine transporter (NET)
 - Once inside, a portion of norepinephrine is re-sequestered into synaptic vesicles via a vesicular monoamine transporter (VMAT)
- Also happens for other small-molecule transmitters

- Clinical importance:

- When reuptake is inhibited, neurotransmitter effects become increased and prolonged
- Many antidepressant drugs act by inhibiting the reuptake of amine transmitters, prolonging their synaptic activity
- Glutamate reuptake into neurons and glia is critical because glutamate is an *excitotoxin*
 - Excess glutamate overstimulates cells, leading to cell death
 - During ischemia or anoxia, impaired glutamate increases, leading to neuronal death

SMALL-MOLECULE TRANSMITTERS: EXCITATORY & INHIBITORY AMINO ACIDS, ACETYLCHOLINE, AND MONOAMINES

GLUTAMATE

- **Primary excitatory neurotransmitter in the brain and spinal cord**

- Responsible for approximately 75% of excitatory transmission in the CNS

- Stored in vesicles by vesicular glutamate transporter (VGLUT)

- Clinical importance:

- Excess glutamate leads to excitotoxicity, which brings neuronal death via Ca overload)
- Memory and learning is affected by NMDA receptor blockade (prevents LTP)
- Metabotropic glutamate receptor 5 (mGluR5) dysregulation is linked to major psychiatric disorders

GABA

- **Major inhibitory neurotransmitter in the brain**

- Mediates both presynaptic and postsynaptic inhibition

- Synthesis:

- Formed from glutamate by the enzyme glutamate decarboxylase (GAD)
- Stored in vesicles by vesicular GABA transporter (VGAT)
- Degraded by GABA transaminase (GABA-T) enzyme

- After release, GABA is rapidly taken back up by a high-affinity GABA transporter by recycling and regulation

- Clinical importance:

- **Dysfunction is linked to sleep disorders, epilepsy, anxiety, and movement disorders** (since it reduces neuronal excitability)

GLYCINE

- **Dual role (only found in CNS)**

- Excitatory
 - Binds to NMDA receptors -> increases glutamate action
 - Glycine spillover into interstitial fluid of **spinal cord dorsal horn to facilitate pain transmission**
- Inhibitory

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- **Major inhibitory neurotransmitter in brainstem and SC**
 - Receptor
 - a subunit - ligand binding
 - b subunit - structural
 - Open Cl channels -> increases Cl conductance (hyperpolarization) -> inhibition

- Glycine balances excitatory and inhibitory signaling in the CNS, especially in the spinal cord

ACETYLCHOLINE

- Released by all neurons exiting the CNS
(technically, PNS)

- Site of transmission

- Neuromuscular junction (skeletal muscle contraction)
- Autonomic ganglia
- Postganglionic parasympathetic junctions: effector organ junctions
- Some postganglionic sympathetic junctions: (e.g., sweat glands)

- Synthesis:

- Formed in nerve terminal from acetyl + acetyl-CoA by choline acetyltransferase (ChAT) enzyme
- Choline reuptake happens by the help of Na-dependent choline transporter (CHT)
- Stored in vesicles by vesicle-associated transported (VAT)
- Released with Ca influx during nerve impulse

- Clinical importance:

- **Involved in neuromuscular control, autonomic regulation, and higher brain functions: (memory, learning, sleep)**
 - Present in basal forebrain complex (projects to hippocampus and neocortex) -> involved in memory, learning, and sleep-wake cycle regulation

ACETYLCHOLINE RECEPTORS

- Two major types

- Nicotinic receptors

- Mechanism: ligand-gated ion channels (ionotropic)
- Subtypes:
 - NMJ - 2a, b, delta, and gamma/epsilon
 - NN - (CNS & autonomic ganglia) - a and b only
- Binding of 2 ACh opens channel, which causes Na influx (depolarization)
- NN receptors are also permeable to Ca, often presynaptic, enhancing glutamate release

- Muscarinic receptors

- Mechanism: G-protein-coupled receptor (metabotropic)
- Subtypes:
 - CNS (M1-5)
 - M1 (autonomic ganglia) - modulate neurotransmission
 - Heart (M2)
 - Glands and smooth muscles (M3)
- Coupled to adenylyl cyclase, K channels, or phospholipase C

NOREPINEPHRINE & EPINEPHRINE

- Released from neurons in the brain

- **Norepinephrine**

- Main transmitter of most sympathetic postganglionic neurons (stored in dense-core vesicles)
- Noradrenergic neurons - neurons containing norepinephrine
 - Located in the locus coeruleus and brainstem nuclei
 - It projects to descend into the spinal cord, to enter cerebellum, and to extend to the thalamus, basal telencephalon, and the entire neocortex
 - Modulate neurotransmission

- **Epinephrine**

- Released in medulla
- Released mainly by adrenal medulla

ADRENORECEPTORS

WEEK 1: INTEGRATIVE PHYSIOLOGY

- α -Adrenoreceptors:

- α_1 - (G-protein G_q , activates phospholipase C (PLC), which bring inositol triphosphate (IP3) & diacylglycerol (DAG) to be made)
 - Excitatory
 - Increases in Ca production and activates protein kinase C
 - Location: smooth muscle, heart
- α_2 (G-protein G_i decrease the production of cAMP, which open K channel, inhibiting Ca transmission)
 - Inhibitory
 - In the presynaptic autoreceptors, it decreases NE release
 - Location: CNS, pancreatic islets, nerve terminals

- β -Adrenoreceptors (**G-protein G_s increases production of cAMP**)

- β_1 : heart, renal juxtaglomerular cells
- β_2 : bronchial & vascular smooth muscle
- β_3 : adipose tissue

- **NE: primary sympathetic neurotransmitter, favors α -receptors**

- **EPI: adrenal medulla hormone, favors β -receptors**

- Both regulate autonomic function + CNS modulation

DOPAMINE

- Dopaminergic pathways:

- **Nigrostriatal system**
 - From substantia nigra, striatum of basal ganglia
 - **Motor control**
- **Mesocortical/mesolimbic pathways**
 - From ventral tegmental area of nucleus accumbens, limbic areas, and cortex
 - **Reward, addiction. psychiatric disorders**

- When aging, loss of dopamine receptors happen

- Receptors

- D1-like family (D1, D5)
 - cAMP excites
- D2-like family (D2-4)
 - cAMP inhibits
 - Overstimulation of D2 receptors - schizophrenia

- D3 - localized in nucleus accumbens

- D4 - high affinity for clozapine (antipsychotic drug)

- Clinical importance:

- Schizophrenia
 - Positive symptoms (hallucinations, delusions, racing thoughts)
 - Negative symptoms (apathy, low motivation, impaired adaptability)
 - Associated with mesocortical dopamine dysfunction
 - Antipsychotic drug blocks D2 receptors (as well as D4)
- Parkinson's disease
 - Low dopamine in nigrostriatal pathway
 - Problem with movement, reward, and cognition

SEROTONIN

- **Highest concentrations are found in blood platelets and GI tract (enterochromaffin cells, myenteric plexus)**

- In the CNS, it is present in raphe nuclei of the brainstem

- Precursor: tryptophan (essential amino acid)
- Its axons project widely to the hypothalamus, limbic system, neocortex, cerebellum, and spinal cord

- Regulates GI function, mood, sleep, pain, and vomiting reflex

- Receptors:

- Most are GPCRs -> acts via adenylyl cyclase or phospholipase C

- Clinical relevance:

- **Depression, anxiety, and GI disorders**

LARGE-MOLECULE TRANSMITTERS: NEUROPEPTIDES

SUBSTANCE P

- Polypeptide with 11 amino acid

- **Acts on neurokinin receptors (NK1-3) - all are GPCRs**

- Preferred ligand for NK1 receptors in CNS
- Happens if second messengers are produced (IP3 and DAG)

WEEK 1: INTEGRATIVE PHYSIOLOGY

- High concentrations in:

- Intestine (role in peristalsis)
- Peripheral nerves
- CNS (afferent neurons in SC dorsal horn)

- Roles:

- Tachykinin neuropeptide crucial for pain transmission, mood regulation, gut motility, and a target for antiemetic & antidepressant drugs
- Mediates first synapse in pain transmission (spinal cord dorsal horn)
- Present in monoamine-rich regions (NORE, SERO, DOPA pathways)
- In hypothalamus - neuroendocrine regulation
- In intestine - gut motility

- Clinical relevance:

- **Substance P - important for chronic pain therapy**
- NK1 receptor - for antidepressant activity; used as antiemetics (especially in chemotherapy)

OPIOID PEPTIDE

- The presence of morphine-binding receptors in the brain and GI tract led to discovery of this endogenous ligands

- Two closely related pentapeptides were first identified:

- Met-enkephalin
- Leu-enkephalin
 - These 2 opioid peptides, along with others, bind to opioid receptors and function as synaptic transmitters

- Enkephalins are found in nerve endings in the GI tract and many regions of the brain

- High concentrations are seen in the substantia gelatinosa, where they exhibit analgesic activity when injected into the brainstem
- Decreased intestinal motility

- **Endogenous opioid peptides (enkephalins, endorphins, dynorphins)**

- **Act via μ , κ , δ (mu, kappa, delta) GPCRs**
- Produce analgesia, reduced gut motility, and mood effects

- Receptors:

- Three main classes: μ (mu), κ (kappa), δ (delta)
 - all are GPCRs (inhibit adenylyl cyclase)
- mu receptor activation
 - Increases K conductance, leading to hyperpolarization of neurons to inhibit firing of AP

- kappa and delta receptor activation
 - Closes Ca channels

OTHER POLYPEPTIDES

1.) Somatostatin

- Found in various parts of the brain, where it acts as a neurotransmitter

- Influences sensory input, locomotor activity, and cognitive function

- In hypothalamus - functions as growth

hormone-inhibiting hormone (GHIH) and is secreted into the portal hypophyseal vessels

- In pancreas - inhibits insulin and other hormone secretion

- In GI tract - important inhibitory regulator

- Receptors:

- **5 somatostatin receptors (SSTR1-SSTR5) - all are GPCRs (inhibit adenylyl cyclase)**
 - SSTR2 - mediates cognitive effects & growth hormone inhibition
 - SSTR5 - mediates insulin inhibition

2.) Vasopressin & Oxytocin

- Function as both hormones

- Secreted by neurons that project to the brainstem and spinal cord

3.) Other neuropeptides in brain and GI tract

- Bradykinin, angiotensin II, endothelin - released by brain neurons

- GI hormones also present in brain:

- Vasoactive intestinal peptide (VIP) - vasodilator in vasomotor nerve fibers

4.) Calcitonin Gene-Related Peptide (CGRP)

- Present in CNS, PNS, GI tract, CVS, urogenital system

- Colocalization: often found with **substance P** or **ACh**

- Causes vasodilation when injected

- CGRP and calcitonin both came from calcitonin gene

- **CGRP - little effect on Ca metabolism**
- Calcitonin - weak vasodilator

- Receptors:

- **Two types of metabotropic CGRP receptors**

- Clinical relevance:

- **Migraine** if released from trigeminal afferent

WEEK 1: INTEGRATIVE PHYSIOLOGY

5.) Neuropeptide Y (NPY)

- Abundant in the brain and ANS

- Functions:

- **Mobilizes Ca and inhibits adenylyl cyclase (GPCRs)**
- In CNS, increases food intake
- In periphery, causes vasoconstriction
- Y1 and Y5 antagonists may help treat obesity
- Acts on heteroreceptors of postganglionic sympathetic nerve terminals to reduce norepinephrine release

guanylyl cyclase for neurotransmission, plasticity, and physiological functions

GAS TRANSMITTERS

1.) Nitric Oxide (NO)

- Originally identified as endothelium-derived relaxing factor released from blood vessels

- Also produced in the brain and ENS

- Freely diffuses across membranes because its a gas.

Thus, it is not stored in vesicles

- Synthesis:

- From **arginine**, it is catalyzed by NO synthase
- Synthesis is triggered by Na receptor activation
-> Ca influx -> neuronal NO synthase activation

- Functions:

- Activates guanylyl cyclase directly
- Diffuses to presynaptic endings to enhance glutamate release
- Presynaptic plasticity, memory, learning
- Acts as retrograde messenger (from post back to pre neuron)

2.) Carbon Monoxide (CO)

- Produced endogenously in CNS neurons and ENS neurons

- Synthesis:

- Formed by heme degradation via heme oxygenase-2

- Functions:

- Stimulates soluble guanylyl cyclase
- Modulates neurotransmission
- Roles in olfaction, pain, long-term potentiation
- May reduce endotoxin-induced vasopressin release centrally

BOTH NO & CO are gaseous, freely diffusable, synthesized when needed, and act mainly through